Sohan Jayram Reddy Kadiri

Seattle | sohanjreddy@outlook.com | (206) 380-6390 | github.com/sohanreddyk | linkedin.com/in/sohanreddyk | www.sohanreddy.org/

PROFESSIONAL SUMMARY

Software Engineer specializing in distributed systems and cloud infrastructure, with experience building backend services using Go, Java, Python and AWS. Skilled in concurrency, fault tolerance, container orchestration and observability.

EDUCATION

Northeastern University Sep 2025 - May 2027
Master of Science, *Computer Science* GPA: 3.9

Siddaganga Institute of Technology Dec 2021 - Jun 2025
Bachelor of Technology, *Artificial Intelligence and Data Science*

PROJECTS

Insight — Distributed Real-Time Analytics Platform | TypeScript, NestJS, Kafka, PostgreSQL, Redis, Docker

- Architected a **multi-tenant distributed event-processing platform** using Kafka, PostgreSQL and Redis, integrating REST APIs, tenant isolation and real-time event delivery.
- Implemented **idempotent Kafka processing and transactional PostgreSQL aggregation**, preventing duplicate counts and isolating failed events through dead-letter queues.
- Increased processing throughput by **61×**, from 46 to 2,816 events/sec, using Kafka batch consumption, bulk PostgreSQL writes and pipelined Redis operations.

Harbor — Distributed LLM Reliability Platform | Go, Redis, Python, FastAPI, Docker, Prometheus, Grafana

- Engineered an **OpenAI-compatible API gateway in Go** with streaming, semantic caching and model routing, exposing Prometheus metrics for observability.
- Implemented **Redis-backed rate limiting, circuit breakers and automatic failover**, handling 400/400 requests during a simulated primary-provider outage.
- Accelerated cached responses by **9.5×** through embedding-based semantic caching, achieving an 85% cache-hit rate under a simulated benchmark workload.

TaskForge — Distributed C++ Task Queue | C++17, Redis, Multithreading, Next.js

- Engineered a **Redis-backed task queue in C++17** with multithreaded workers, priority scheduling and at-least-once delivery, processing 5,000 tasks/sec.
- Implemented **configurable retries and stuck-task recovery**, achieving 99.9% successful task completion and recovery within 5 seconds during fault-injection tests.
- Optimized **concurrent task execution and worker scheduling**, achieving 20 ms p95 dispatch latency across 16 concurrent workers.

WORK EXPERIENCE

Siemens | Software Engineer Intern | Remote Aug 2024 - Jul 2025

- Engineered **Node.js backend services and PostgreSQL-backed REST APIs** across 2 production applications, serving 2,500+ users with 250 ms p95 API latency.
- Integrated a **Pinecone-backed RAG pipeline** into an AI chatbot handling 1,200+ conversations, improving retrieval relevance by 40%.
- **Led 3 engineering interns** through architecture design and API integration, resolving 9 production issues through debugging, log analysis and service-level testing.
- Automated **Docker-based deployments using Azure and Vercel CI/CD**, reducing release time by 70%, from 20 to 6 minutes.

Unisys | Cloud Engineering Intern | Remote May 2024 - Jan 2025

- Developed **RESTful backend services** for a digital marketplace serving 8,000+ users and processing 1,600+ orders, generating $20K in gross merchandise value.
- Engineered a **WebSocket-based messaging system** supporting 350 concurrent connections, sub-150 ms message latency and 99.8% delivery reliability.
- Optimized **checkout APIs and transaction workflows** through one-click checkout, increasing conversion by 22% and contributing $14K in revenue.
- Improved **mobile application performance** through Lighthouse-guided optimizations, reducing page load time to under 1.2 seconds and increasing mobile conversion by 8%.

SKILLS

Languages: Go, Java, Python, C++, TypeScript, JavaScript, SQL

Systems & Backend: Distributed Systems, Concurrency, Multithreading, REST APIs, WebSockets, Microservices

Cloud & Infrastructure: AWS (EC2, S3, Lambda), Azure, Docker, Kubernetes, Terraform, Git, GitHub Actions, CI/CD

Databases & Messaging: PostgreSQL, Redis, MongoDB, Kafka, RabbitMQ

Observability & Reliability: Prometheus, Grafana, OpenTelemetry, Distributed Tracing, Rate Limiting, Circuit Breakers

CERTIFICATIONS & ACHIEVEMENTS

- Open Source - Merged PR #472 into mindsdb/anton, fixing a bug that deleted saved user rules and adding 3 regression tests. Sep 2026
- Publication - "Detection of DDoS Attacks in Fog Computing using Interpretable Quantum-based 2Q Learning Approach" - IEEE Jan 2025